

Supplementary Online Content

Writing Group for the PReVENT Investigators. Effect of a low vs intermediate tidal volumes strategy on ventilator-free days in intensive care unit patients without ARDS: a randomized clinical trial. *JAMA*. doi:10.1001/jama.2018.14280

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The authors, to give readers additional information about their work, have provided this supplementary material.

eMETHODS

Ventilation protocol

The allowed ventilation modes were volume-controlled ventilation and pressure support ventilation (pressure-controlled ventilation, modes volume combining controlled and spontaneous ventilation, and automated modes of ventilation were not allowed). The inspiration-to-expiration ratio with volume-controlled ventilation was 1:2. With volume-controlled ventilation the inspiration time and pause were set at 25% and 10% respectively. With pressure support ventilation the highest possible pressure rise was chosen, and cycling off was set at 25%.

The inspired oxygen fraction was 0.21 or higher to maintain oxygen saturation 90 to 92% and/or $\text{PaO}_2 > 7.3$ to 10.7 kPa (55 to 80 mmHg). The respiratory rate was to be adjusted to maintain an arterial pH of 7.25 to 7.45. In case of a metabolic acidosis or – alkalosis, a lower or higher than normal PaCO_2 was to be accepted, left to the discretion of the attending physician. The lowest level of positive end–expiratory pressure was 5 cm H_2O ; allowed FiO_2 –PEEP combinations according to previous used strategy.¹ Recruitment maneuvers were allowed, when deemed necessary by the attending physician.

Standard Care

Standard care followed local clinical guidelines and was performed by independent board-certified ICU physicians and board-certified ICU nurses not involved in the trial. The nurse-to-patient ratio in the participating centers varied from 1:1 to 1:2, depending on severity of illness. ICU physicians rotated in 24/7 shifts, meaning that there was always an ICU physician present in the ICU. Physician and nursing staffs used restrictive sedation, preferring analgo-sedation over hypno-sedation, and restrictive intravenous fluid resuscitation. ICU nurses performed standard airway care, including

endotracheal suctioning, and tracheostomy care if applicable. Endotracheal suction was performed only when clinically indicated and according to current guidelines.² All patients received humidification (passive or active) of the inhaled air.

Infection prevention

Infection prevention strategies consisted of frequent oral care (tooth brushing and rinsing of the oral cavity every six hours), head-of-bed elevation, hand washing, and use of selective decontamination of the digestive tract.³

eREFERENCES

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3. Oostdijk EAN, Kesecioglu J, Schultz MJ, et al. Notice of Retraction and Replacement: Oostdijk et al. Effects of Decontamination of the Oropharynx and Intestinal Tract on Antibiotic Resistance in ICUs: A Randomized Clinical Trial. *JAMA*. 2014;312(14):1429-1437. *JAMA*. 2017;317(15):1583. doi: 10.1001/jama.2017.1282.

eTable 1 – Respiratory variables during the first three days of ventilation

	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value
Number of patients	452	463		391	410		321	343		272	296	
V _T , mL/kg PBW	5.9 (5.2 – 6.7)	9.1 (7.9 – 10.0)	< 0.001	6.6 (5.5 – 8.7)	9.3 (8.1 – 10.1)	< 0.001	7.4 (5.8 – 9.7)	9.1 (7.7 – 10.3)	< 0.001	7.3 (5.9 – 9.1)	9.1 (7.7 – 10.5)	< 0.001
Respiratory rate, bpm	23 (19 – 28)	18 (14 – 20)	< 0.001	22 (16 – 28)	16 (13 – 21)	< 0.001	20 (16 – 27)	17 (13 – 22)	< 0.001	21 (15 – 27)	18 (14 – 24)	< 0.001
MV, L/min	9.2 (7.5 – 10.9)	10.0 (8.5 – 12.3)	< 0.001	9.9 (8.2 – 12.0)	9.8 (8.1 – 12.0)	0.99	10.0 (8.3 – 12.0)	9.9 (8.0 – 12.0)	0.45	10.1 (8.2 – 12.2)	10.6 (8.6 – 13.2)	0.14
Plateau pressure, cmH ₂ O	18 (15 – 21)	21 (18 – 24)	< 0.001	17 (14 – 20)	20 (18 – 23)	< 0.001	18 (16 – 22)	21 (17 – 24)	0.02	20 (16 – 24)	22 (18 – 23)	0.18
PEEP, cmH ₂ O	8 (5 – 10)	8 (5 – 8)	0.33	6 (5 – 8)	7 (5 – 8)	0.64	6 (5 – 8)	6 (5 – 8)	0.58	7 (5 – 8)	6 (5 – 8)	0.29
Driving pressure, cmH ₂ O	10 (8 – 13)	14 (12 – 16)	< 0.001	10 (8 – 13)	13 (11 – 15)	< 0.001	10 (8 – 13)	14 (11 – 16)	< 0.001	11 (8 – 14)	14 (12 – 16)	< 0.001
FiO ₂	0.45 (0.37 – 0.55)	0.40 (0.40 – 0.60)	0.66	0.38 (0.30 – 0.45)	0.35 (0.30 – 0.40)	0.02	0.35 (0.30 – 0.45)	0.35 (0.30 – 0.45)	0.94	0.35 (0.30 – 0.40)	0.35 (0.30 – 0.45)	0.83
PaO ₂ / FiO ₂ , mmHg	210 (152 – 300)	214 (153 – 307)	0.49	230 (171 – 300)	243 (180 – 311)	0.21	225 (178 – 292)	240 (176 – 297)	0.62	236 (181 – 285)	235 (168 – 307)	0.89
PaCO ₂ , mmHg	43.5 (38.2 – 48.7)	36.0 (30.7 – 42.0)	< 0.001	39.7 (35.2 – 44.2)	35.2 (31.5 – 39.7)	< 0.001	39.7 (36.0 – 45.0)	37.5 (33.7 – 42.0)	< 0.001	41.9 (36.0 – 45.7)	38.0 (33.7 – 42.7)	< 0.001
pHa	7.31 (7.24 – 7.54)	7.39 (7.31 – 7.46)	< 0.001	7.38 (7.33 – 7.42)	7.43 (7.37 – 7.48)	< 0.001	7.41 (7.36 – 7.45)	7.43 (7.38 – 7.47)	< 0.001	7.44 (7.39 – 7.48)	7.45 (7.40 – 7.48)	0.02
pHa < 7.25	96 / 381 (25.2)	46 / 404 (11.4)	< 0.001	18 / 375 (4.8)	14 / 393 (3.6)	0.40	2 / 281 (0.7)	3 / 303 (1.0)	0.71	7 / 237 (3.0)	1 / 248 (0.4)	0.03
Respiratory acidosis*	132 / 381 (34.6)	49 / 404 (12.1)	< 0.001	42 / 375 (11.2)	21 / 393 (5.3)	0.003	29 / 281 (10.3)	21 / 303 (6.9)	0.14	17 / 237 (7.2)	8 / 248 (3.2)	0.05

Data are N (%) median (interquartile range)
V_T: tidal volume; PBW: predicted body weight; bpm: breaths per minute; MV: minute ventilation; PEEP: positive end-expiratory pressure; FiO₂: fraction of inspired oxygen; PaO₂: arterial oxygen pressure; PaCO₂: arterial pressure of carbon dioxide; pHa: arterial pH
* defined as pHa < 7.35 and PaCO₂ > 45 mmHg
The lower number of patients assessed for respiratory acidosis and pHa < 7.25 represent missing in ABG measurements

eTable 2 – Ventilator settings during the first three days of ventilation

Volume-Controlled Ventilation												
	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low	Intermediate	p	Low	Intermediate	p	Low	Intermediate	p	Low	Intermediate	p
	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value
Number of patients	309	362		129	214		61	68		46	42	
V _T , mL/kg PBW	5.8 (5.0 – 6.2)	9.3 (8.1 – 10.1)	< 0.001	5.5 (4.7 – 6.1)	9.5 (8.5 – 10.1)	< 0.001	5.5 (4.9 – 6.0)	9.3 (8.0 – 9.9)	< 0.001	5.8 (5.2 – 6.4)	9.5 (8.5 – 9.9)	< 0.001
Respiratory rate, bpm	25 (20 – 28)	18 (14 – 20)	< 0.001	28 (22 – 30)	16 (12 – 20)	< 0.001	30 (23 – 32)	16 (12 – 22)	< 0.001	27 (24 – 32)	15 (12 – 20)	< 0.001
MV, L/min	9.0 (7.6 – 10.8)	10.2 (8.7 – 12.3)	< 0.001	9.7 (8.3 – 11.8)	9.6 (8.2 – 11.4)	0.49	10.0 (8.4 – 12.1)	9.8 (8.1 – 11.6)	0.43	10.2 (8.3 – 12.7)	8.9 (7.7 – 11.9)	0.07
Plateau pressure, cmH ₂ O	18 (15 – 21)	21 (18 – 24)	< 0.001	17 (14 – 20)	20 (18 – 23)	< 0.001	19 (17 – 22)	21 (17 – 24)	0.04	20 (17 – 23)	22 (18 – 23)	0.20
PEEP, cmH ₂ O	8 (5 – 10)	8 (5 – 8)	0.11	7 (5 – 8)	7 (5 – 8)	0.36	8 (5 – 10)	7 (5 – 8)	0.25	8 (5 – 10)	6 (5 – 8)	0.08
Driving pressure, cmH ₂ O	10 (8 – 13)	14 (12 – 16)	< 0.001	10 (8 – 13)	13 (11 – 15)	< 0.001	10 (8 – 13)	14 (11 – 16)	< 0.001	11 (9 – 14)	14 (12 – 16)	0.001
Pressure Support Ventilation												
	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low	Intermediate	p	Low	Intermediate	p	Low	Intermediate	p	Low	Intermediate	p
	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value
Number of patients	110	73		225	164		211	223		181	192	
V _T , mL/kg PBW	7.5 (6.1 – 8.7)	8.7 (7.3 – 9.8)	< 0.001	7.8 (6.3 – 9.5)	9.0 (7.6 – 10.3)	< 0.001	8.3 (6.5 – 10.3)	9.0 (7.6 – 10.5)	0.03	7.9 (6.5 – 9.7)	9.0 (7.7 – 10.7)	< 0.001
Respiratory rate, bpm	20 (16 – 24)	17 (14 – 21)	0.01	19 (14 – 24)	17 (13 – 22)	0.03	18 (14 – 23)	17 (13 – 22)	0.03	19 (15 – 25)	18 (14 – 24)	0.28
MV, L/min	9.3 (7.7 – 11.6)	9.2 (7.6 – 12.4)	0.76	9.8 (7.9 – 11.9)	10.2 (8.1 – 12.40)	0.35	10.1 (8.3 – 12.0)	10.0 (8.0 – 12.1)	0.56	10.0 (8.1 – 11.9)	10.9 (8.9 – 13.6)	0.01
Maximum pressure, cmH ₂ O	17 (14 – 21)	21 (17 – 25)	< 0.001	15 (12 – 20)	20 (16 – 25)	< 0.001	15 (11 – 19)	19 (15 – 22)	< 0.001	15 (12 – 19)	18 (15 – 22)	< 0.001
PEEP, cmH ₂ O	7 (5 – 8)	6 (5 – 8)	0.85	6 (5 – 8)	8 (5 – 8)	0.68	6 (5 – 8)	5 (5 – 8)	0.68	6 (5 – 8)	6 (5 – 8)	0.88
Pressure support, cmH ₂ O	11 (7 – 14)	14 (11 – 18)	< 0.001	8 (6 – 11)	14 (10 – 16)	< 0.001	7 (6 – 11)	12 (8 – 15)	< 0.001	8 (6 – 12)	11 (8 – 15)	< 0.001
Other Mode of Ventilation [#]												
	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low	Intermediate	p	Low	Intermediate	p	Low	Intermediate	p	Low	Intermediate	p
	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value	Tidal Volume	Tidal Volume	value
Number of patients	33	28		37	32		49	52		45	62	
V _T , mL/kg PBW	5.9 (5.2 – 7.6)	6.9 (5.6 – 8.5)	0.17	7.2 (6.2 – 8.5)	7.9 (6.6 – 8.7)	0.34	6.5 (5.8 – 7.6)	8.8 (8.2 – 9.6)	0.04	8.2 (7.8 – 8.8)	8.6 (7.0 – 9.4)	0.94
Respiratory rate, bpm	20 (17 – 24)	19 (15 – 26)	0.97	20 (16 – 27)	21 (12 – 28)	0.62	21 (17 – 25)	20 (15 – 21)	0.46	19 (15 – 26)	22 (20 – 26)	0.27
MV, L/min	9.1 (6.4 – 10.2)	9.7 (7.3 – 12.7)	0.08	10.7 (8.7 – 14.8)	10.5 (7.9 – 14.1)	0.85	10.0 (7.9 – 11.1)	11.0 (8.2 – 12.4)	0.42	13.1 (10.9 – 15.6)	11.0 (9.5 – 12.8)	0.83
Plateau pressure (or maximum pressure),* cmH ₂ O	19 (14 – 19)	23 (22 – 24)	0.11	17 (17 – 17)	21 (17 – 24)	0.50	26 (22 – 31)	29 (25 – 33)	0.04	25 (23 – 31)	30 (26 – 33)	0.03
PEEP, cmH ₂ O	5 (5 – 10)	8 (5 – 1)	0.50	8 (5 – 10)	8 (5 – 8)	0.54	5 (5 – 12)	10 (6 – 12)	0.37	13 (12 – 15)	6 (5 – 8)	0.05
Driving pressure, cmH ₂ O	10 (9 – 12)	14 (11 – 16)	0.43	11 (10 – 11)	12 (12 – 16)	0.37	17 (15 – 22)	21 (18 – 25)	0.001	19 (15 – 21)	22 (19 – 26)	< 0.001

Data are N (%) and median (interquartile range); [#] modes other than allowed in the protocol and including pressure controlled ventilation, Neurally Adjusted Ventilator Assist and Adaptive Support Ventilation; * maximum pressure when plateau pressure was not available with the used mode
V_T: tidal volume; *PBW*: predicted body weight; *bpm*: breaths per minute; *MV*: minute ventilation; *PEEP*: positive end-expiratory pressure;

eTable 3 – Respiratory variables during the first three days of ventilation

Volume-Controlled Ventilation												
	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value
Number of patients	309	362		129	214		61	68		46	42	
FiO ₂	0.45 (0.40 – 0.60)	0.40 (0.40 – 0.60)	0.18	0.40 (0.30 – 0.50)	0.35 (0.30 – 0.40)	0.001	0.40 (0.35 – 0.45)	0.35 (0.30 – 0.42)	0.06	0.40 (0.35 – 0.45)	0.35 (0.25 – 0.48)	0.28
PaO ₂ / FiO ₂ , mmHg	199 (149 – 287)	204 (151 – 312)	0.24	221 (164 – 283)	245 (180 – 317)	0.03	210 (169 – 274)	240 (172 – 296)	0.26	202 (158 – 247)	249 (154 – 318)	0.22
PaCO ₂ , mmHg	43.5 (39.0 – 49.5)	35.2 (30.0 – 40.5)	< 0.001	41.2 (37.5 – 46.0)	33.7 (30.7 – 37.5)	< 0.001	42.0 (38.2 – 47.2)	36.0 (33.0 – 39.0)	< 0.001	43.5 (38.2 – 49.3)	37.5 (35.4 – 41.1)	< 0.001
Arterial pH	7.30 (7.23 – 7.36)	7.40 (7.32 – 7.48)	< 0.001	7.35 (7.30 – 7.40)	7.43 (7.36 – 7.48)	< 0.001	7.38 (7.32 – 7.42)	7.43 (7.40 – 7.47)	< 0.001	7.39 (7.35 – 7.43)	7.43 (7.39 – 7.47)	0.007
Arterial pH < 7.25	74 / 264 (27.9)	33 / 319 (10.3)	< 0.001	11 / 129 (8.5)	8 / 213 (3.8)	0.07	0 / 59 (0.0)	0 / 67 (0.0)	---	3 / 46 (6.5)	0 / 42 (0.0)	0.28
Respiratory acidosis*	100 / 264 (37.9)	33 / 319 (10.3)	< 0.001	21 / 129 (16.3)	10 / 213 (4.7)	< 0.001	9 / 59 (15.3)	2 / 67 (3.0)	0.01	6 / 46 (13.0)	2 / 42 (4.8)	0.18
Pressure Support Ventilation												
	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value
Number of patients	110	73		225	164		211	223		181	192	
FiO ₂	0.40 (0.35 – 0.50)	0.40 (0.35 – 0.50)	0.51	0.35 (0.30 – 0.45)	0.35 (0.30 – 0.40)	0.53	0.35 (0.30 – 0.40)	0.35 (0.30 – 0.40)	0.79	0.35 (0.30 – 0.40)	0.35 (0.30 – 0.40)	0.36
PaO ₂ / FiO ₂ , mmHg	239 (168 – 315)	240 (179 – 293)	0.70	237 (182 – 309)	240 (187 – 306)	0.96	227 (184 – 294)	245 (183 – 300)	0.57	241 (190 – 290)	228 (172 – 304)	0.52
PaCO ₂ , mmHg	42.7 (37.5 – 47.2)	40.5 (36.0 – 44.2)	0.23	39.0 (35.2 – 43.5)	37.5 (33.6 – 42.0)	0.04	39.0 (36.0 – 44.2)	37.5 (33.7 – 42.9)	0.03	39.7 (35.8 – 45.0)	38.2 (33.0 – 42.7)	0.01
Arterial pH	7.35 (7.27 – 7.40)	7.35 (7.29 – 7.40)	0.78	7.40 (7.35 – 7.43)	7.43 (7.37 – 7.46)	< 0.001	7.42 (7.38 – 7.46)	7.43 (7.38 – 7.47)	0.27	7.45 (7.40 – 7.48)	7.45 (7.41 – 7.49)	0.42
Arterial pH < 7.25	18 / 97 (18.6)	8 / 65 (12.3)	0.29	6 / 221 (2.7)	6 / 161 (3.8)	0.57	2 / 204 (1.0)	3 / 218 (1.4)	0.70	2 / 180 (1.1)	1 / 189 (0.5)	0.53
Respiratory acidosis*	25 / 97 (25.8)	13 / 65 (20.0)	0.40	18 / 221 (8.1)	10 / 161 (6.2)	0.47	17 / 204 (8.3)	17 / 218 (7.8)	0.84	9 / 180 (5.0)	5 / 189 (2.6)	0.24
Other Mode of Ventilation [#]												
	After titration on day of randomization			Day 1			Day 2			Day 3		
	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value	Low Tidal Volume	Intermediate Tidal Volume	p value
Number of patients	33	28		37	32		49	52		45	62	
FiO ₂	0.50 (0.40 – 0.60)	0.50 (0.40 – 0.60)	0.87	0.35 (0.25 – 0.60)	0.30 (0.25 – 0.50)	0.50	0.37 (0.26 – 0.48)	0.50 (0.40 – 0.70)	0.02	0.35 (0.25 – 0.55)	0.37 (0.30 – 0.45)	0.66
PaO ₂ / FiO ₂ , mmHg	211 (156 – 273)	219 (170 – 363)	0.68	239 (153 – 349)	270 (135 – 334)	0.90	244 (135 – 368)	157 (108 – 272)	0.07	225 (157 – 277)	219 (164 – 292)	0.75
PaCO ₂ , mmHg	43.9 (37.3 – 48.7)	39.7 (33.7 – 42.9)	0.08	36.0 (33.7 – 40.5)	38.2 (30.4 – 42.4)	0.72	40.5 (36.4 – 44.6)	38.6 (33.9 – 41.7)	0.26	36.0 (33.7 – 39.4)	39.0 (35.2 – 45.0)	0.33
Arterial pH	7.32 (7.27 – 7.37)	7.38 (7.30 – 7.42)	0.12	7.37 (7.32 – 7.41)	7.45 (7.38 – 7.48)	0.02	7.39 (7.36 – 7.41)	7.44 (7.37 – 7.47)	0.26	7.41 (7.36 – 7.49)	7.46 (7.42 – 7.49)	0.31
Arterial pH < 7.25	4 / 20 (20.0)	5 / 20 (25.0)	0.99	1 / 25 (4.0)	0 / 19 (0.0)	0.99	0 / 18 (0.0)	0 / 18 (0.0)	---	2 / 11 (18.2)	0 / 17 (0.0)	0.15
Respiratory acidosis*	7 / 20 (35.)	3 / 20 (15.0)	0.27	3 / 25 (12.0)	1 / 19 (5.3)	0.62	3 / 18 (16.7)	2 / 18 (11.1)	0.99	2 / 11 (18.2)	1 / 17 (5.9)	0.54

Data are number / total (%) or median (interquartile range); [#] modes other than allowed in the protocol and including pressure controlled ventilation, Neurally Adjusted Ventilator Assist and Adaptive Support Ventilation;

FiO₂: fraction of inspired oxygen; PaO₂: arterial oxygen pressure; PaCO₂: arterial pressure of carbon dioxide; pH_a: arterial pH

* defined as pH_a < 7.35 and PaCO₂ > 45 mmHg

The lower number of patients assessed for respiratory acidosis and pH_a < 7.25 represent missing in ABG measurements

eTable 4 – Sedatives, analgesics, neuromuscular blockade, fluids, vasopressors and co-interventions during the first three days

	Low Tidal Volume (n = 477)	Intermediate Tidal Volume (n = 484)	p value
Sedative infusion**	320 (70.6)	333 (72.1)	0.63
Days of sedative administration	0.6 (0.3 – 1.5)	0.6 (0.0 – 1.5)	0.76
Analgesic infusion***	277 (61.1)	273 (59.1)	0.53
Days of analgesic administration	0.7 (0.3 – 2.0)	0.7 (0.4 – 2.0)	0.65
Lowest RASS			
Day 1	-4 (-5 to -3)	-4 (-5 to -3)	0.58
Day 2	-4 (-4 to -1)	-3 (-4 to -1)	0.17
Day 3	-3 (-4 to -1)	-3 (-4 to -1)	0.01
Neuromuscular blockade£	53 (11.7)	60 (13.0)	0.55
Days of neuromuscular blockade	0.0 (0.0 – 0.3)	0.0 (0.0 – 0.3)	0.55
Fluid balance in 24 hours, mL			
Day 1	1225 (-18 – 2471)	1310 (108 – 2632)	0.42
Day 2	259 (-869 – 1288)	167 (-998 – 1203)	0.59
Day 3	-278 (-1619 – 570)	-530 (-1932 – 641)	0.39
Use of colloids*	100 / 454 (22.0)	107 / 464 (23.1)	0.71
Volume of colloids	0 (0 – 0)	0 (0 – 0)	0.97
Volume of crystalloids	6892 (3531 – 10643)	7161 (3828 – 11054)	0.50
Transfusion of red blood cells	129 / 454 (28.4)	121 / 464 (26.1)	0.43
Units of red blood cells	0 (0 – 1)	0 (0 – 1)	0.50
Transfusion of fresh frozen plasma	30 / 454 (6.6)	40 / 464 (8.6)	0.25
Units of fresh frozen plasma	0 (0 – 0)	0 (0 – 0)	0.24
Transfusion of platelets	48 / 454 (10.6)	39 / 464 (8.4)	0.26
Units of platelets	0 (0 – 0)	0 (0 – 0)	0.26
Use of vasopressors ^a	363 (80.0)	353 (76.4)	0.19
Days of vasopressor use	2 (1 – 4)	2 (1 – 4)	0.87
Use of rescue therapy			
Decreasing of instrumental dead space	4 (0.8%)	1 (0.2%)	0.21
Recruitment maneuvers	30 (6.3%)	25 (5.2%)	0.45
Prone positioning	22 (4.6%)	24 (5.0%)	0.80
Inhaled nitric oxide	1 (0.2%)	1 (0.2%)	1.00
High frequency oscillatory ventilation	0 (0.0%)	0 (0.0%)	NA
Extracorporeal membrane oxygenation	1 (0.2%)	0 (0.0%)	0.49
Extracorporeal removal of carbon dioxide	0 (0.0%)	0 (0.0%)	NA

Data are number / total (%) or median (interquartile range)

RASS: *Richmond-Agitation Sedation Scale*;

^a defined as need of any dose of norepinephrine and/or epinephrine, and dopamine > 5 µg/kg/min

* defined as Hydroxyethyl starch, gelatin, dextran or albumin

** defined as any infusion for more than one hour of clonidine, dexmedetomidine, midazolam, lorazepam, diazepam, propofol and/or thiopental

*** defined as any infusion for more than one hour of fentanyl, ketamine, methadone, morphine, remifentanyl and/or sufentanyl

£ defined as any infusion for more than one hour of atracurium, cisatracurium, mevacurium, pancuronium, rocuronium and, or vecuronium

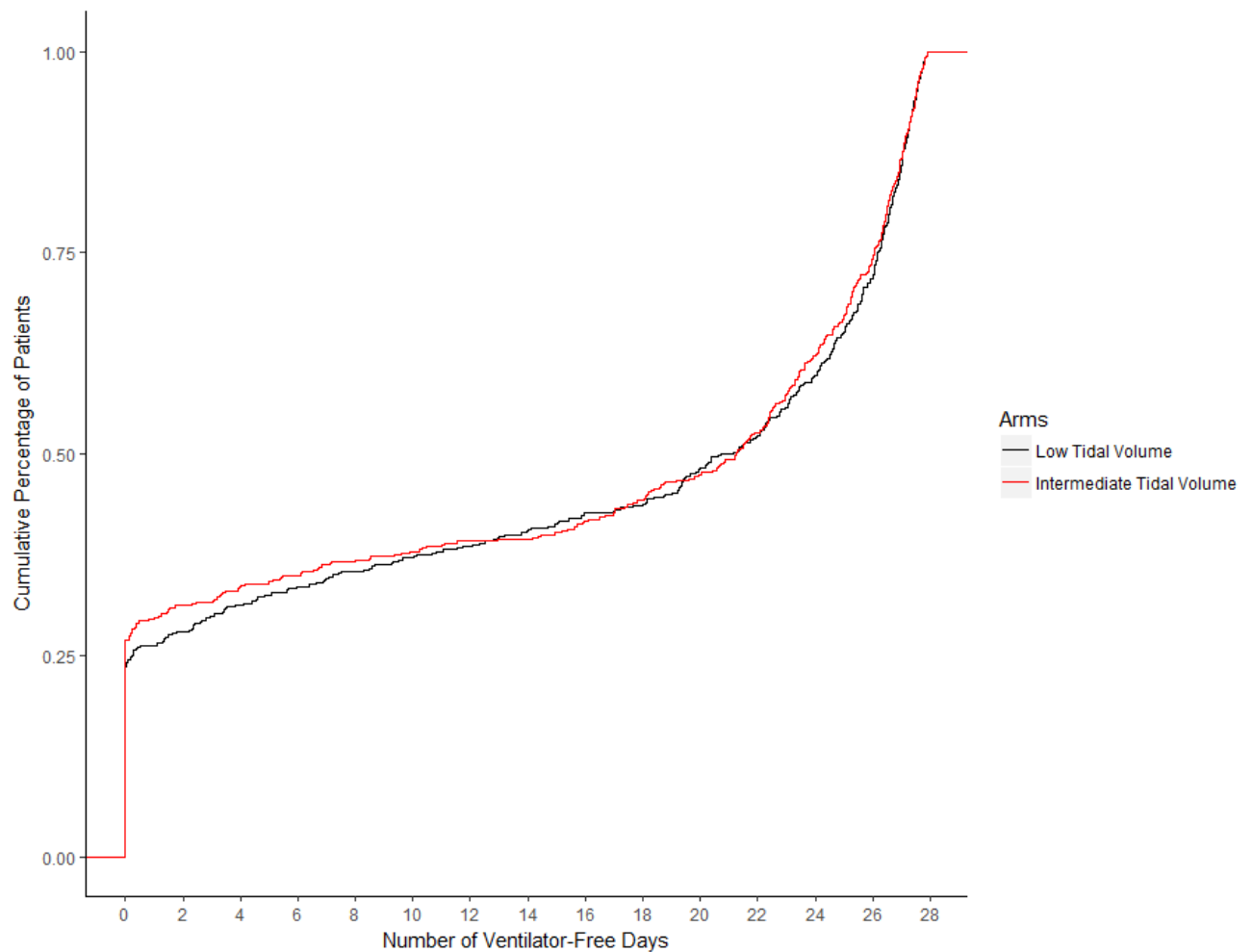
eTable 5 – Subgroup effects

Subgroup	Low Tidal Volume	Intermediate Tidal Volume	Mean Difference (95% CI)	<i>p</i> for interaction
Reason for ventilation				
Pneumonia	14.0 ± 11.4	13.1 ± 11.2	0.86 (-2.48 – 4.20)	0.16
Non-Pneumonia	15.4 ± 11.6	16.0 ± 11.5	-0.50 (-2.13 – 1.13)	
Sepsis				
Yes	12.0 ± 11.7	14.2 ± 10.8	-2.20 (-5.46 – 1.06)	0.17
No	16.0 ± 11.4	15.7 ± 11.6	0.25 (-1.38 – 1.89)	
PaO ₂ / FiO ₂				
≤ 200	14.6 ± 11.5	15.5 ± 11.3	-0.87 (-3.21 – 1.47)	0.32
> 200	15.7 ± 11.9	14.8 ± 11.8	0.87 (-1.62 – 3.36)	
LIPS				
≥ 4	14.0 ± 11.4	14.2 ± 11.4	-0.19 (-2.06 – 1.66)	0.96
< 4	17.0 ± 11.6	17.3 ± 11.3	-0.27 (-2.61 – 2.07)	
Reason for admission				
Clinical	14.5 ± 11.7	15.2 ± 11.5	-0.70 (-2.32 – 0.92)	0.23
Surgical	18.7 ± 10.3	17.0 ± 11.1	1.70 (-1.62 – 5.03)	
SAPS II				
≥ 50	12.0 ± 11.8	12.8 ± 11.9	-0.88 (-2.96 – 1.18)	0.26
< 50	19.5 ± 9.7	18.8 ± 10.0	0.75 (-1.14 – 2.64)	
Ventilation before randomization				
> 6 hours	16.5 ± 11.6	16.8 ± 10.7	-0.29 (-7.04 – 6.45)	0.98
≤ 6 hours	15.2 ± 11.6	15.4 ± 11.5	-0.21 (-1.72 – 1.28)	
Start of ventilation				
ICU	12.7 ± 11.2	15.2 ± 11.0	-2.50 (-4.63 – -0.36)	0.01
Outside ICU	17.2 ± 11.5	15.8 ± 11.8	1.45 (-0.52 – 3.43)	

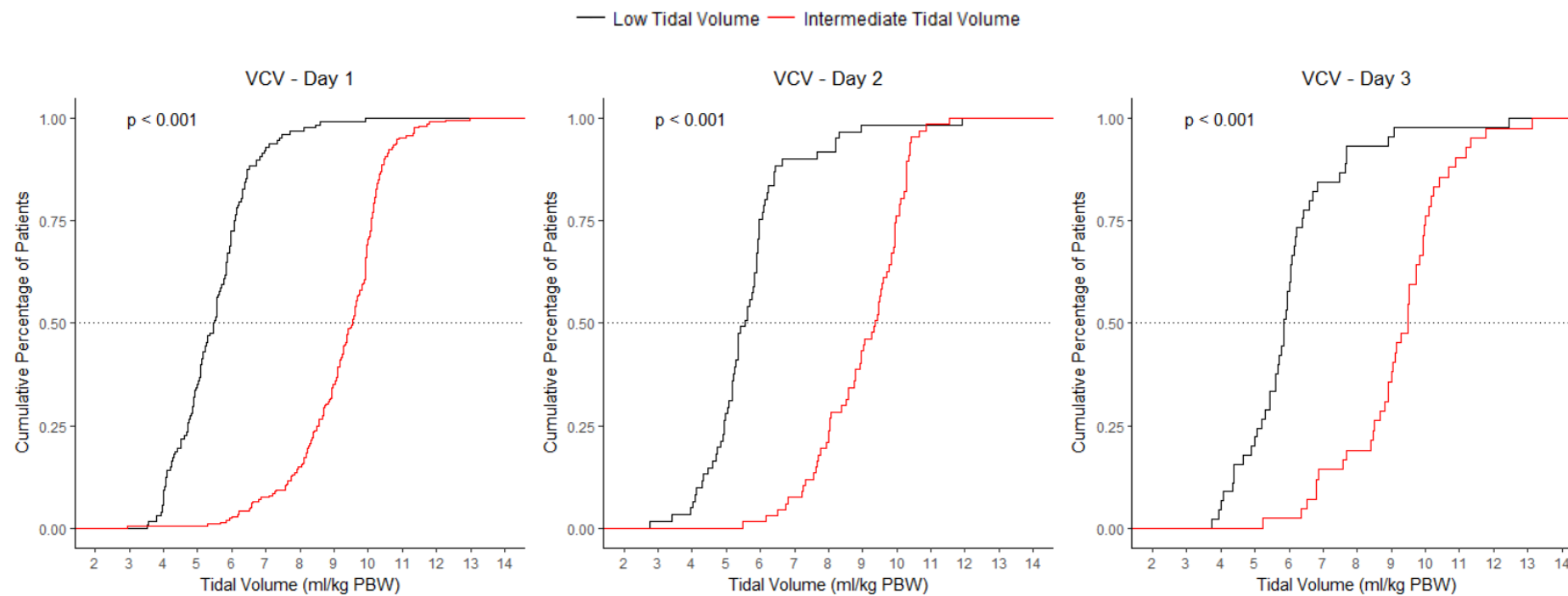
Data are mean (SD)

PaO₂ / FiO₂: arterial partial oxygen pressure / fraction of inspired oxygen; LIPS: Lung Injury Prediction Score; SAPS: Simplified Acute Physiology Score; ICU: Intensive care unit; bpm: breaths per minute

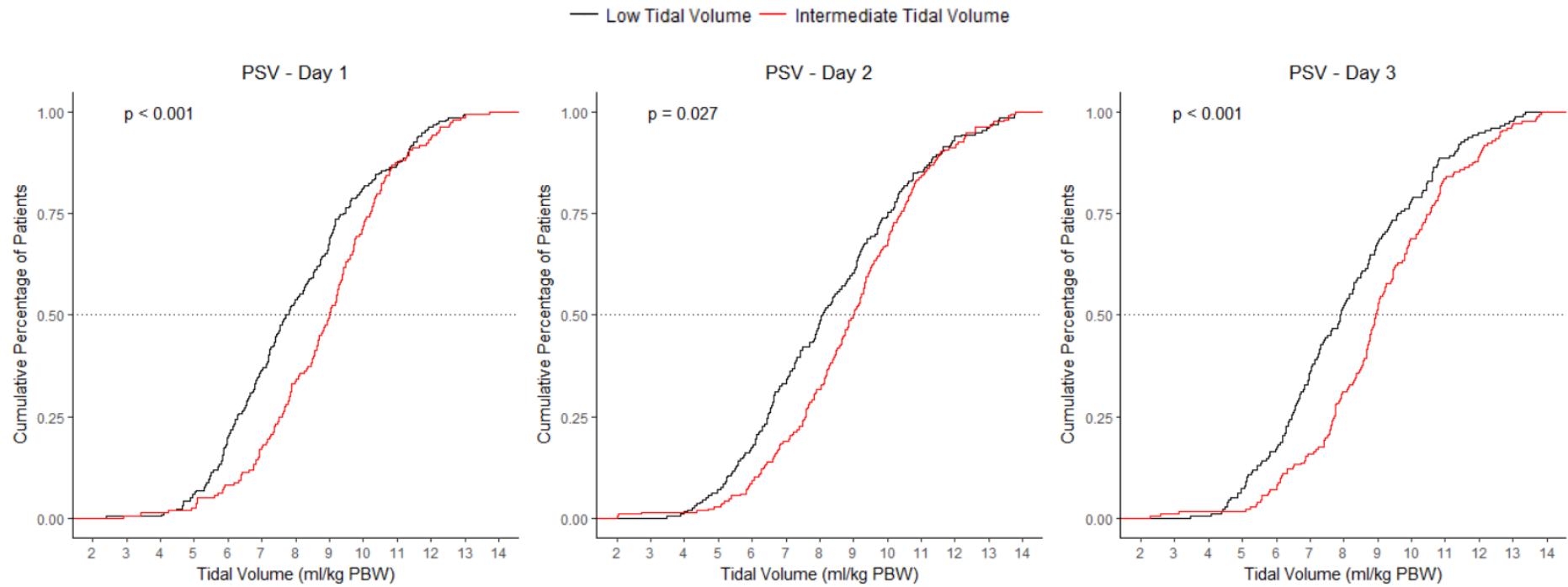
eFigure 1. Ventilator-free days and cumulative proportion of patients free from ventilation



eFigure 2. Distribution of tidal volumes in volume controlled ventilation



eFigure 3. Distribution of tidal volumes in pressure support ventilation



eFigure 4. Distribution of pressure support level in pressure support ventilation

